eBPF-driven Fine-grained Telemetry and Machine Learning for EVPN/VXLAN Datacenter Networks

Pre-Study

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1. Introduction

Modern datacenter networks are expected to support both high throughput and low latency for a large variety of applications. In recent years, EVPN and VXLAN have been widely adopted in datacenter environments because they provide scalable and flexible network virtualization for multi-tenant systems [1], [2]. However, as datacenter traffic becomes increasingly dynamic, short periods of concentrated traffic may create transient congestion and cause sudden performance degradation.

One important example of such transient congestion is the microburst. A microburst refers to a very short but intense burst of packets arriving within a limited period of time. Although the duration of a microburst is short, it may lead to queue buildup, packet loss, and significant taillatency increase [3]. For latency-sensitive services, these short-term spikes may still have a noticeable impact on application performance. Therefore, understanding and detecting these events is an important problem in datacenter networking research.

Traditional monitoring approaches are often not sufficient for this task, because they usually provide only coarse-grained measurements and may fail to capture short-lived changes before latency spikes occur. In contrast, eBPF makes it possible to collect fine-grained telemetry directly from the Linux kernel datapath with relatively low overhead [4]. This creates an opportunity to study whether kernel-level metrics can provide useful signals before congestion becomes visible at the application level.

Based on this background, this pre-study investigates the theoretical and methodological foundation of a master's degree project on eBPF-driven telemetry and simple machine learning methods for EVPN/VXLAN datacenter networks. The goal is not to fully solve latency prediction in production datacenters, but rather to examine whether a software-based testbed can be used to reproduce latency spikes and to study whether kernel-level telemetry may help detect them at an early stage. The purpose of this pre-study is therefore to review relevant background, discuss the choice of methods, and outline a feasible research design for the project.

2. Background and Related Work

Datacenter networks are commonly built using Spine–Leaf architectures, where Leaf switches connect to hosts and Spine switches provide connectivity across the network fabric. In such environments, overlay technologies are often used to separate logical tenant networks from the underlying physical infrastructure. VXLAN provides a widely used mechanism for encapsulating Layer 2 traffic over Layer 3 networks, while EVPN provides a BGP-based control plane for distributing reachability information in a scalable way [1], [2]. As a result, EVPN/VXLAN has become a representative design for modern virtualized datacenter networks.

Within this type of environment, network performance is not determined only by average throughput or average delay. Short-lived congestion events may still cause serious performance problems. Previous work has shown that microbursts can create rapid queue buildup and may lead to packet drops and latency increase even when average utilization is not extremely high [3]. This is one reason why tail latency, rather than only average latency, is often used as an important performance metric in datacenter studies. Recent work has also highlighted that the

estimation and understanding of tail latency remain challenging, especially under bursty traffic conditions [5].

To understand these phenomena, many studies have focused on network telemetry and observability. Existing approaches include programmable dataplane telemetry, in-band network telemetry, and software-based end-host monitoring [6]. These methods differ in deployment complexity, visibility, and overhead. Hardware-assisted telemetry may provide detailed information from network devices, but such solutions are not always accessible in small research projects. Software-based methods are often more practical for experimental environments, although they may not capture every part of the network state.

In this context, eBPF has attracted increasing attention as a flexible mechanism for monitoring and analyzing system and network behavior. It can be used to attach lightweight programs to different kernel events and packet-processing paths, which makes it suitable for collecting packet-related and queue-related indicators in Linux-based systems [4]. Recent studies on endhost diagnosis and host-side traffic behavior also suggest that important performance signals can sometimes be observed from the host or kernel side, rather than only from dedicated network hardware [7], [8]. This is particularly relevant for this project, since the main interest is to study whether kernel-level telemetry can provide early signs of latency spikes in a software-based EVPN/VXLAN testbed.

Overall, the literature suggests three important points. First, microbursts and short-lived congestion can have a meaningful impact on tail latency. Second, fine-grained telemetry is necessary if such short events are to be observed. Third, eBPF and host-side monitoring provide a realistic and relatively accessible way to study these issues in experimental environments. At the same time, current work still leaves room for studying how kernel-level signals and latency spikes can be connected in a controlled EVPN/VXLAN software testbed.

3. Method Choice and Research Design

The main methodological choice in this project is to use a software-based testbed instead of relying on data from a production datacenter. This choice is mainly motivated by feasibility and controllability. Access to real datacenter traces, hardware telemetry, or production network devices is usually limited for student projects. In contrast, a software-based environment allows traffic patterns, topology, and congestion conditions to be adjusted in a controlled and repeatable manner. For a master's degree project, this makes it possible to carry out experiments within a realistic but manageable scope.

A second important choice is to use EVPN/VXLAN as the target network environment. This choice is not intended to reproduce all the complexity of a production cloud network, but rather to provide a meaningful overlay-network context for the study. Since EVPN/VXLAN is widely used in modern datacenters, using it in the testbed helps make the project more relevant than a purely generic queueing experiment. At the same time, it remains feasible to emulate such an environment with Linux network namespaces and FRR.

The third key choice is to collect telemetry using eBPF. Compared with hardware-based monitoring, eBPF is more accessible and easier to integrate into a software testbed. It also allows fine-grained observations directly from the Linux kernel datapath. In this project, the goal is not to claim that eBPF can observe all network-internal behavior, but to examine whether

observable kernel-level indicators, such as packet rate, drop counters, and queue-related signals, may change before latency spikes become visible at the application level.

The research design of the project follows a relatively simple logic. First, a small EVPN/VXLAN testbed will be constructed in a Linux-based environment. Second, bursty traffic patterns such as incast traffic, short-flow bursts, or competing background flows will be generated in order to create temporary congestion. Third, telemetry data will be collected through eBPF, while end-to-end latency measurements will be used as ground truth. Fourth, the collected data will be processed using sliding time windows so that telemetry signals can be aligned with latency behavior. Finally, simple prediction models or threshold-based methods will be evaluated to determine whether they can identify latency spikes shortly before they occur.

The use of simple models is also a deliberate methodological choice. Since this project is mainly a feasibility-oriented study, the main interest is not to achieve the best possible prediction accuracy, but to examine whether there is a meaningful relationship between kernel-level telemetry and latency spikes. Simpler models are easier to interpret, require less data, and are more suitable for a small-scale experiment. This makes them appropriate for a master's degree project in which the emphasis is on understanding the problem and demonstrating a realistic proof of concept.

4. Expected Challenges and Limitations

Although the proposed research design is feasible, it also has several limitations. First, the project will be conducted in a small-scale software-based testbed. Such an environment is useful for controlled experiments, but it cannot fully represent the scale, traffic diversity, and hardware behavior of a production datacenter. Therefore, the conclusions of the study should be interpreted mainly as evidence of feasibility rather than as direct proof of deployment effectiveness in real-world networks.

Second, the telemetry collected in this project will be limited to kernel-level and host-visible signals. While these signals may still be useful, they do not provide complete visibility into switch-internal buffers or all network-wide dynamics. As a result, some causes of latency spikes may remain only partially observable in the experiment. This should be recognized as an inherent limitation of a host-side monitoring approach.

Third, the reproduction of microbursts in an experimental setting may not be identical to how such events occur in production datacenters. In the testbed, microbursts will likely be generated through controlled traffic patterns such as incast or bursty short flows. This is a reasonable approach for experimentation, but it still means that the study focuses on representative rather than naturally occurring production scenarios.

Another challenge concerns data quantity and model complexity. Since the testbed will be relatively small, the amount of data that can be generated may also be limited. For this reason, it is more realistic to focus on simple and interpretable prediction methods rather than complex machine learning models. This choice supports the feasibility of the project, but it also means that the study will not aim for state-of-the-art predictive performance.

Finally, the implementation itself may involve practical difficulties. Building an EVPN/VXLAN testbed, generating repeatable congestion patterns, attaching eBPF programs, and synchronizing telemetry with latency measurements all require careful setup. These tasks are achievable, but they may affect the time available for deeper analysis. Therefore, it is

important to keep the scope of the project realistic and to prioritize the most essential research questions.

5. Conclusion

This pre-study has reviewed the basic background and methodological foundation of a project on eBPF-driven fine-grained telemetry for EVPN/VXLAN datacenter networks. The discussion shows that microbursts and short-lived congestion are relevant problems because they may have a strong impact on tail latency, while traditional monitoring methods are often too coarse to observe such events in time. The literature also suggests that eBPF provides a useful and practical way to collect fine-grained host-side telemetry in Linux-based environments.

Based on these observations, the proposed project chooses a software-based EVPN/VXLAN testbed, eBPF-based telemetry, and simple prediction methods as its main approach. This combination is suitable for a master's degree project because it balances research relevance with practical feasibility. Although the study will not fully represent the complexity of production datacenters, it can still provide useful insights into whether kernel-level signals may help identify latency spikes at an early stage.

In summary, the pre-study indicates that the project is both meaningful and feasible within a limited research scope. The next step is to implement the testbed, generate controlled bursty traffic, collect telemetry and latency data, and evaluate whether the observed signals have predictive value. These activities will provide the basis for the later stages of the degree project.

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